# Peiyang Song

GHC 6601, 5000 Forbes Avenue, Pittsburgh, PA 15213
✉ psong2@andrew.cmu.edu
peiyang-song.github.io

## Education

8/2026 – now **Carnegie Mellon University** Pittsburgh, PA
*Ph.D. Student in Artificial Intelligence @ Language Technologies Institute (LTI)*
Advised by Prof. Sean Welleck. Also working closely with Prof. Jeremy Avigad.

6/2026 **California Institute of Technology** Pasadena, CA
*B.S. in Computer Science & Minor in Robotics*
Advised by Prof. Steven Low & Prof. Günter Niemeyer. GPA: **4.2/4.0**.

## Research Interests

Formal Reasoning · Natural Language Reasoning · Neuro-Symbolic AI · AI4MATH · Agents

## Selected Publications & Preprints

Full list on Google Scholar: `https://scholar.google.com/citations?user=E1j11NQAAAAJ`

Under Review **LLM2Cert: LLM Synthesis of Certificates with SMT/Lean Verification**
Zhipeng Cao*, Peiyang Song*, Nikolaus Vertovec, Weiru Cui, Zhihao Yang, Mingkang Liu, Luxia Lin, Bai Xue†, Alessandro Abate† (* Equal Contribution, † Equal Advising)
*arXiv Preprint, 2026*

Technical Report **Épi: Towards a Persistent, Self-Improving Agent for Autonomous Research**
Yue Huang, Zhangchen Xu, Xiaonan Luo, Zheyuan Liu, Lichi Li, Yiping Wang, Peiyang Song, Eric Gabriel Ramos, Enze Xie, Olexandr Isayev, Paul Liang, Nouha Dziri, Huan Sun, Jianfeng Gao, Ying Liu, Misha Sra, Xiangliang Zhang, Alex Pentland, Zichen Chen
*Technical Report, 2026*

NeurIPS 2026 **P$^3$: Joint Program-and-Proof Planning for Verified Code Generation**
Zenan Li*, Ziran Yang*, Peiyang Song, Zhaoyu Li, Kaiyu Yang (* Equal Contribution)
*Neural Information Processing Systems (NeurIPS), 2026*

NeurIPS 2026 **Vero: Can AI Agents Build Formally Verified Software Repositories?**
Zhe Ye*, Hantao Lou*, Yuechun Sun*, Peiyang Song, Zhengxu Yan, Timothe Kasriel, Qingyang Zhang, Kaiyu Yang, Soonho Kong, Jingxuan He, Dawn Song (* Equal Contribution)
*Neural Information Processing Systems (NeurIPS), 2026*

NeurIPS 2026 **Rethinking Psychometric Evaluation of LLMs: When and Why Self-Reports Predict Behavior**

Rafal D. Kocielnik, Pengrui Han, Peiyang Song, Myrl G. Marmarelis, Ramit Debnath, Dean Mobbs, Anima Anandkumar, R. Michael Alvarez

*Neural Information Processing Systems (NeurIPS), 2026*, **Best Paper Award @ ICML CTB Workshop**

Under Review **Interactive Evaluation Requires a Design Science**

Keyang Xuan*, Peiyang Song*, Pan Lu, Pengrui Han, Wenkai Li, Zhenyu Zhang, Zexue He, Wenyue Hua, Manling Li, Jiaxuan You, Adrian Weller, Yizhong Wang†, and Jiaxin Pei† (* Equal Contribution, † Equal Advising)

*ICML Workshop on Combining Theory and Benchmarks (CTB), 2026*

Under Review **Human Agency in Initial Human-AI Proof Formalization Workflows**

Katherine M. Collins*, Simon Frieder*, Jonas Bayer, Jacob Loader, Jeck Lim, Peiyang Song, Fabian Zasier, Lexin Zhou, Shanda Li, Sam Looi, Jose Hernandez-Orallo, Joshua B. Tenenbaum, Cameron Freer, Umang Bhatt, Adrian Weller, Valerie Chen†, Ilia Sucholutsky† (* Equal Contribution, † Equal Advising)

*NeurIPS Workshop on Mathematical Reasoning and AI (MATH-AI), 2025*

Under Review **Steer2Adapt: Dynamically Composing Steering Vectors Elicits Efficient Adaptation of LLMs**

Pengrui Han*, Xueqiang Xu*, Keyang Xuan*, Peiyang Song, Siru Ouyang, Runchu Tian, Yuqing Jiang, Qian Cheng, Pengcheng Jiang, Jiashuo Sun, Junxia Cui, Ming Zhong, Ge Liu, Jiawei Han, Jiaxuan You (* Equal Contribution)

*ICLR Workshop on Representational Alignment (Re-Align), 2026*

TMLR 2026 **Large Language Model Reasoning Failures**

Peiyang Song*, Pengrui Han*, Noah Goodman (* Equal Contribution)

*Transactions on Machine Learning Research, 2026*, **Survey Certification**

ICML 2026 **The Personality Illusion: Revealing Dissociation Between Self-Reports & Behavior in LLMs**

Pengrui Han*, Rafal D. Kocielnik*, Peiyang Song, Ramit Debnath, Dean Mobbs, Anima Anandkumar, R. Michael Alvarez (* Equal Contribution)

*International Conference on Machine Learning, 2026*

**Best Paper Honorable Mention** @ *NeurIPS LAW Workshop, 2025*

NeuS 2025 **Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean**

Peiyang Song, Kaiyu Yang, Anima Anandkumar

*International Conference on Neuro-symbolic Systems, 2025*

**1.3k+ stars on Github, ranking 2nd after Mathlib4 among all Lean projects**

IEEE Micro 2025 **Delay Space Arithmetic and Architecture**

Rhys Gretsch, Peiyang Song, Advait Madhavan, Jeremy Lau, Timothy Sherwood

*IEEE Micro, 2025*, **Top Pick Award**

ICLR 2025 **LeanAgent: Lifelong Learning for Formal Theorem Proving**

Adarsh Kumarappan*, Mo Tiwari*, Peiyang Song, Robert Joseph George, Chaowei Xiao, Anima Anandkumar

*International Conference on Learning Representations, 2025*

TMLR 2025 **LeanProgress: Guiding Search for Neural Theorem Proving via Proof Progress Prediction**

Robert Joseph George, Suozhi Huang, Peiyang Song, Anima Anandkumar

*Transactions on Machine Learning Research, 2025*

In Submission **Adaptation of Agentic AI: A Survey of Post-Training, Memory, and Skills**

Pengcheng Jiang*, Jiacheng Lin*, Zhiyi Shi*, Zifeng Wang, Luxi He, Yichen Wu, Ming Zhong, Peiyang Song, Qizheng Zhang, Heng Wang, Xueqiang Xu, Hanwen Xu, Pengrui Han, Dylan Zhang, Jiashuo Sun, Chaoqi Yang, Kun Qian, Tian Wang, Changran Hu, Manling Li, Quanzheng Li, Hao Peng, Sheng Wang, Jingbo Shang, Chao Zhang, Jiaxuan You, Liyuan Liu, Pan Lu, Yu Zhang, Heng Ji, Yejin Choi, Dawn Song, Jimeng Sun, Jiawei Han (* Equal Contribution)

*arXiv Preprint, 2025*

Under Review **How and Why LLMs Generalize: A Fine-Grained Analysis of LLM Reasoning from Cognitive Behaviors to Low-Level Patterns**

Haoyue Bai, Yiyou Sun, Wenjie Hu, Shi Qiu, Maggie Ziyu Huan, Peiyang Song, Robert Nowak, Dawn Song

*arXiv Preprint, 2025*

EMNLP 2024 **In-Context Learning May Not Elicit Trustworthy Reasoning: A-Not-B Errors in Pretrained Language Models**

Pengrui Han*, Peiyang Song*, Haofei Yu, Jiaxuan You (* Equal Contribution)

*Conference on Empirical Methods in Natural Language Processing: Findings, 2024*

EMNLP 2024 **Creative and Context-Aware Translation of East Asian Idioms with GPT-4**

Kenan Tang*, Peiyang Song*, Yao Qin, Xifeng Yan (* Equal Contribution)

*Conference on Empirical Methods in Natural Language Processing: Findings, 2024*

ASPLOS 2024 **Energy Efficient Convolution with Temporal Arithmetic**

Rhys Gretsch, Peiyang Song, Advait Madhavan, Jeremy Lau, Timothy Sherwood

*ACM International Conference on Architectural Support for Programming Languages and Operating Systems, 2024*

NeurIPS 2023 **LeanDojo: Theorem Proving with Retrieval-Augmented Language Models**

Kaiyu Yang, Aidan Swope, Alex Gu, Rahul Chalamala, Peiyang Song, Shixing Yu, Saad Godil, Ryan Prenger, Anima Anandkumar

*Conference on Neural Information Processing Systems, 2023*, **Oral Presentation**

## Selected Academic Awards

7/2026 **Best Paper Award @ ICML CTB Workshop**

For paper "Rethinking Psychometric Evaluation of LLMs: When and Why Self-Reports Predict Behavior"

5/2026 **Outstanding Reviewer Award @ EACL**

In recognition of outstanding review service at EACL 2026

1/2026 **Survey Certification @ TMLR**

For paper "Large Language Model Reasoning Failures"

12/2025 **Best Paper Honorable Mention Award @ NeurIPS LAW Workshop**

For paper "The Personality Illusion: Revealing Dissociation Between Self-Reports & Behavior in LLMs"

5/2025 **ICLR Notable Reviewer Award**

In recognition of outstanding review service at ICLR 2025

4/2025 **George W. Housner Student Discovery Fund**

For paper "Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean"

2/2025 **IEEE Micro Top Pick Award**

For paper "Energy Efficient Convolutions with Temporal Arithmetic"

4/2023 **Caltech SURF Fellowship**

For research on machine learning for theorem proving in Lean

## Selected Media

2026 **Scientists Found AI's Fatal Flaw—The Most Advanced Models Are Failing Basic Logic Tests**

*Popular Mechanics*

2026 **'Not how you build a digital mind': How reasoning failures are preventing AI models from achieving human-level intelligence**

*Live Science*

2026 **When AI gets the maths right but the model wrong**

*Doing The Math*

2025 **Researchers Discover "Personality Illusion" to Reveal a Profound Disconnect Between Language and Behavior in LLMs**

*MIT Technology Review China*

2025 **New Framework Simplifies the Complex Landscape of Agentic AI**

*VentureBeat*

2025 **This AI Paper Explains Why Most "Agentic AI" Systems Feel Impressive in Demos and then Completely Fall Apart in Real Use**

*MarkTechPost*

2024 **Mathematicians' Newest Assistants Are Artificially Intelligent**

*Scientific American*

2024 **LeanAgent: The First Life-Long Learning Agent for Formal Theorem Proving in Lean**

*MarkTechPost*

2024 **Lean Copilot: An AI Tool That Allows Large Language Models (LLMs) to Be Used in Lean for Proof Automation**

*MarkTechPost*

2023 **Can LLMs Generate Mathematical Proofs That Can Be Rigorously Checked?**

*MarkTechPost*

## Invited Talks

**From Reasoning Failures to Reliable Intelligence: Understanding, Evaluating, and Building Better LLM Reasoning Systems**

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|---|---|---|
| 8/2026 | XINGHE Talk @ Shanghai AI Lab | Host: Yafu Li |
| 3/2026 | Fireside Chat @ Ploutos.ai | Host: Cecilia Tamura |
| 3/2026 | BlueDot AI Safety Reading Group | Host: Mark Keavney |

## Teaching Experience

Fall 2026 **CS 4824 / ECE 4424: Machine Learning**
Guest Lecturer @ Virginia Tech

Spring 2026 **CS 38/138: Algorithms**
Teaching Assistant @ *Caltech*

Winter 2026 **ME/CS/EE 133B: Robotics – Planning**
Teaching Assistant @ *Caltech*

Fall 2025 **ME/CS/EE 133A: Robotics – Kinematics**
Teaching Assistant @ *Caltech*

## Academic Services

**Lead Organizer** The 6th Workshop on Mathematical Reasoning and AI (MATH-AI) @ NeurIPS 2026

**Lead Instructor** Tutorial on Neuro-Symbolic Theorem Proving with Lean @ 3rd Neuro-Symbolic AI Summer School (NSSS)

**Reviewer** ACM Computing Surveys
Intelligent Computing (Science Partner Journal)

Conference on Neural Information Processing Systems (NeurIPS)
International Conference on Learning Representations (ICLR)
Conference on Language Modeling (COLM)
Annual Meeting of the Association for Computational Linguistics (ACL)
Conference on Empirical Methods in Natural Language Processing (EMNLP)
Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)
European Chapter of the Association for Computational Linguistics (EACL)

Asia-Pacific Chapter of the Association for Computational Linguistics (AACL)
International Joint Conference on Natural Language Processing (IJCNLP)

NeurIPS Mathematical Reasoning and AI (MATH-AI) Workshop
NeurIPS Workshop on Deep Learning for Code (DL4C)
NeurIPS Workshop on Behavioral Machine Learning
ICLR VerifAI: AI Verification in the Wild Workshop
ICLR Workshop on Representational Alignment (Re-Align)
ICML AI for Math (AI4MATH) Workshop
ICML Workshop on LLMs and Cognition (LLM-Cognition)
ICML Workshop on Assessing World Models
ICML Workshop on on Models of Human Feedback for AI Alignment (MoFA)
COLM Workshop on Agent Behavior (WAB)

## On-Campus Services & Appointments

**Caltech** Admissions Ambassador @ Caltech Undergraduate Admissions Office
First-Year Caltech Connector (FCC) @ Student & Family Engagement Office